

5G RAN

CMAS Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-06-28).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between CMAS and RF module fast deep dormancy. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

None

Editorial Changes

Revised descriptions of function impacts. For details, see [4.3.2.3 Function Impacts](#).

Revised descriptions of impact analysis. For details, see [4.2.2 Impacts](#).

Revised descriptions of the PWS restart indication procedure and PWS failure indication procedure in the principles. For details, see [4.1.5 PWS Restart Indication Procedure](#) and [4.1.6 PWS Failure Indication Procedure](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-051104	Public Warning System (PWS)	4 CMAS

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
CMAS	Supported only in SA networking	4 CMAS

2.4 Differences Between High Frequency Bands and Low Frequency Bands

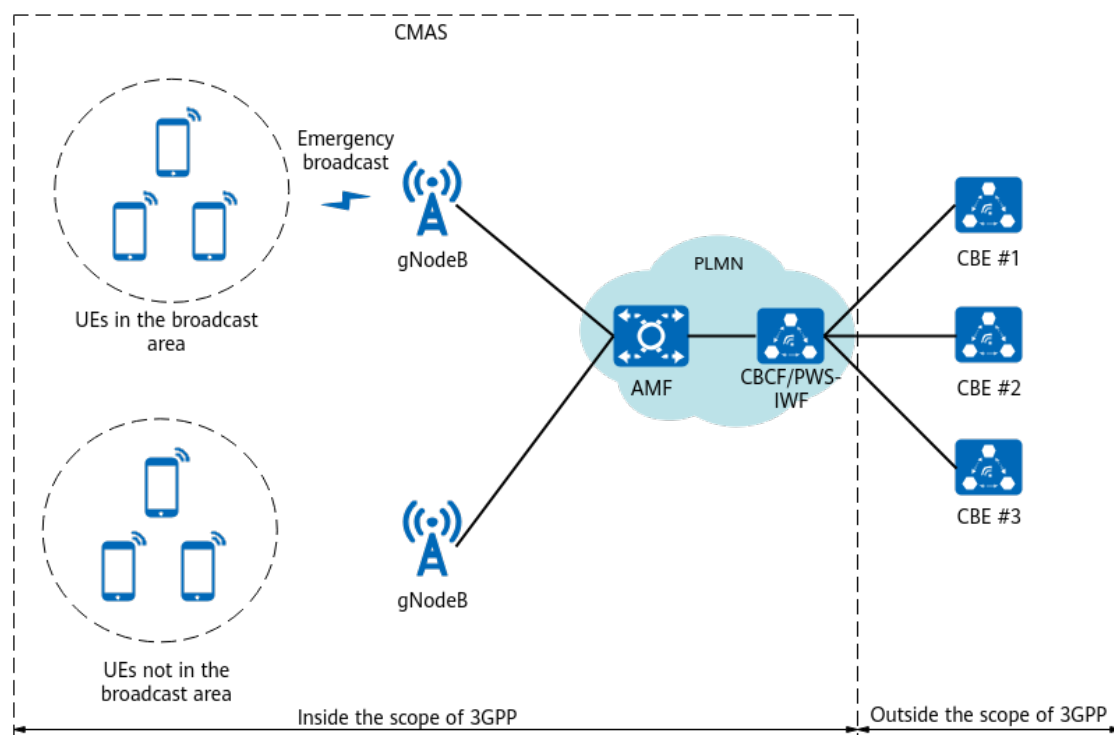
This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
CMAS	Supported only in low frequency bands	4 CMAS

3 Overview

Commercial mobile alert system (CMAS) enables governments or other authorities to broadcast warning notifications about emergencies to the public through public land mobile networks (PLMNs). As an example, CMAS may be used by an authority figure to give an emergency speech, or to broadcast alerts about an imminent terrorist attack or an abducted child.

Figure 3-1 CMAS network architecture



CMAS involves UEs, gNodeBs, the access and mobility management function (AMF), and the cell broadcast center function/public warning system interworking function (CBCF/PWS-IWF) and complies with 3GPP specifications.

A cell broadcast entity (CBE) refers to an organization that provides warning notifications. Communications between CBEs and the CBCF/PWS-IWF are beyond the scope of 3GPP specifications.

Table 3-1 NE functions

NE	Function Description
UE	Receives warning notifications in either idle or connected mode. The UE must support CMAS.
gNodeB	Processes CMAS signaling messages. For details, see section 8.9 in 3GPP TS 38.413.
AMF	Processes CMAS signaling messages over the NG interface. For details, see section 6 in 3GPP TS 23.041.
CBCF/PWS-IWF	Converts and distributes signaling messages to and from a PLMN and a non-PLMN network. For details, see section 5 in 3GPP TS 23.041.
CBE	Sends the warning notifications to the PLMN.

4 CMAS

4.1 Principles

3GPP specifications stipulate the CMAS warning notification specifications. Operators can plan CMAS broadcast areas as required. After the CMAS function is enabled, the CBE initiates a broadcast procedure in the corresponding broadcast area if it detects any potential danger. When the danger is relieved, the CBE initiates a broadcast stop procedure. After PWS restart indication and PWS failure indication are enabled, the gNodeB notifies the AMF of cell exceptions in the broadcast area.

4.1.1 Warning Notification Specifications

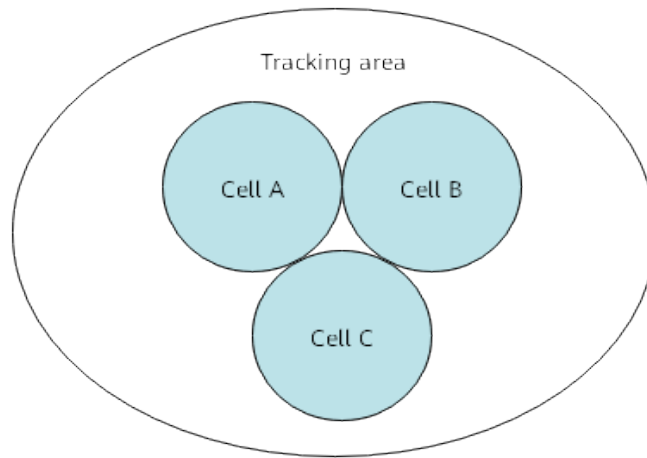
As stipulated in 3GPP TS 38.413, the maximum length of a warning notification is 9600 bytes, as this is the maximum bearer capability of the NG interface. A warning notification can be divided into a maximum of 64 segments.

Information about the warning notifications and their segments is carried using SIB8.

4.1.2 Broadcast Area

A broadcast area specifies the scope within which warning notifications are broadcast. Broadcast areas can be planned by cell or tracking area. Typically, a tracking area covers a larger area than a cell, as shown in [Figure 4-1](#).

Figure 4-1 Broadcast area



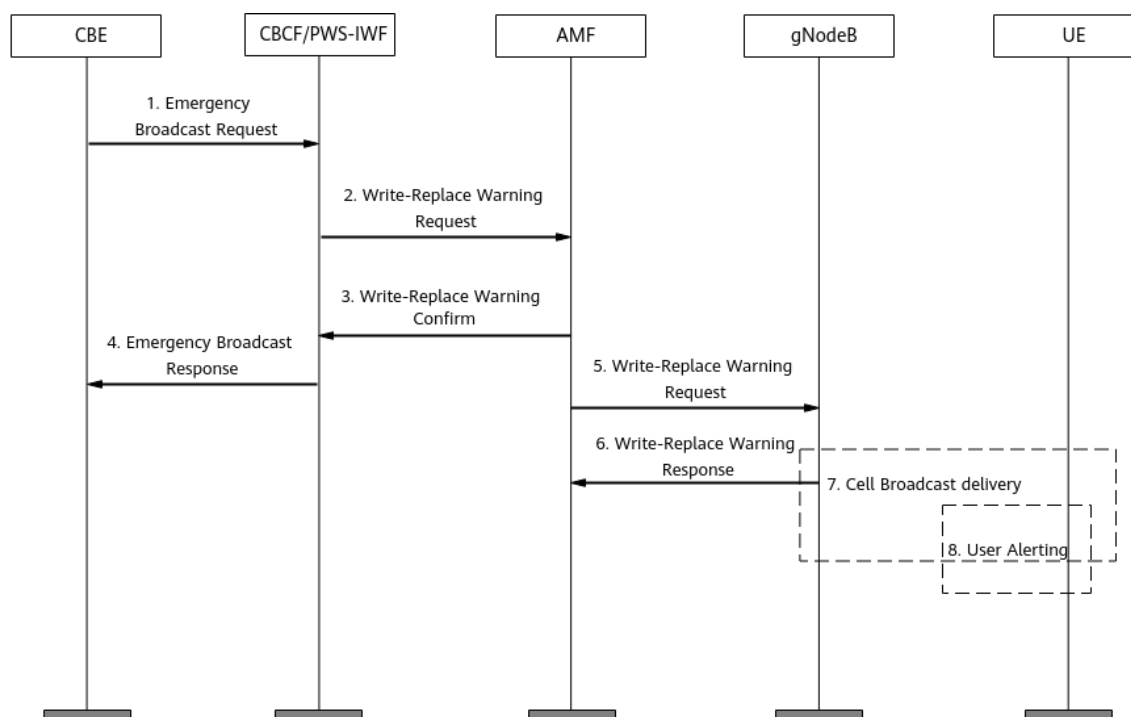
The principles for planning broadcast areas are as follows:

- No additional configuration is required for the operator.
- It is recommended that broadcast areas be planned by tracking area because:
 - A single cell provides relatively small coverage. A list of cells may meet the coverage requirement of a broadcast area, but the cell list needs to be determined on the core network side, which is difficult to implement.
 - A tracking area is more suitable for a broadcast area, in terms of coverage scope, maintenance, and project implementation.

4.1.3 Broadcast Procedure

4.1.3.1 Overall Procedure

CMAS is enabled when the **CMAS_SWITCH** option of the NRDUCellFeatureSw.*PwsSwitch* parameter is selected, the **gNBSibConfig.SibType** parameter is set to **SIB_TYPE_8**, and the **gNBSibConfig.SibTransPolicy** parameter is set to **BROADCAST**. The **gNBSibConfig.SibTransPeriod** parameter specifies the system information transmission period. [Figure 4-2](#) shows the procedure for broadcasting a warning notification.

Figure 4-2 Warning notification broadcast procedure

1. After detecting a potential danger, the CBE determines to broadcast a warning notification to the public through the PLMN and then sends an Emergency Broadcast Request message to the CBCF/PWS-IWF.
2. The CBCF/PWS-IWF converts the received message into a standardized signaling message Write-Replace Warning Request defined by 3GPP TS 23.041 and then sends the message to the AMF.
3. The AMF responds to the CBCF/PWS-IWF with a Write-Replace Warning Confirm message, declaring that it has started forwarding the warning notification to the gNodeB.
4. The CBCF/PWS-IWF responds to the CBE with an Emergency Broadcast Response message, declaring that warning notification broadcasting has started.
5. The AMF forwards the Write-Replace Warning Request message to the gNodeBs within the broadcast area indicated by the *Warning Area List* IE in this message.

The maximum size of a Write-Replace Warning Request message that can be processed by the NG interface of a gNodeB is 64 KB.

NOTE

The Write-Replace Warning Request message is a CMAS warning notification if it contains a *Concurrent Warning Message Indicator* IE. Otherwise, it is an earthquake and tsunami warning system (ETWS) warning notification.

6. The gNodeB determines whether the served cells are located in a broadcast area indicated by the *Warning Area List* IE in the Write-Replace Warning Request message and replies to the AMF with a Write-Replace Warning Response message that contains the information of the areas where the warning notification is to be broadcast. Only warning notifications from the

PLMN whose **gNBOperator.CmasBroadcastSwitch** is set to **ON** can be broadcast over the air interface. Warning notifications from the PLMN whose **gNBOperator.CmasBroadcastSwitch** is set to **OFF** cannot be broadcast over the air interface. The gNodeB determines which PLMN does the CMAS warning notification sent by the AMF belong to based on the operator information and *Warning Area List* after the establishment of an NG link.

The gNodeB can identify the *Warning Area Coordinates* IE in the Write-Replace Warning Request message. This IE contains the coordinates of the warning area related to the warning notification. The value of this IE is a character string with a maximum of 1024 bytes. For details, see 3GPP TS 38.413. In this case, SIB8 contains the *warningAreaCoordinatesSegment* IE. For details, see 3GPP TS 38.331.

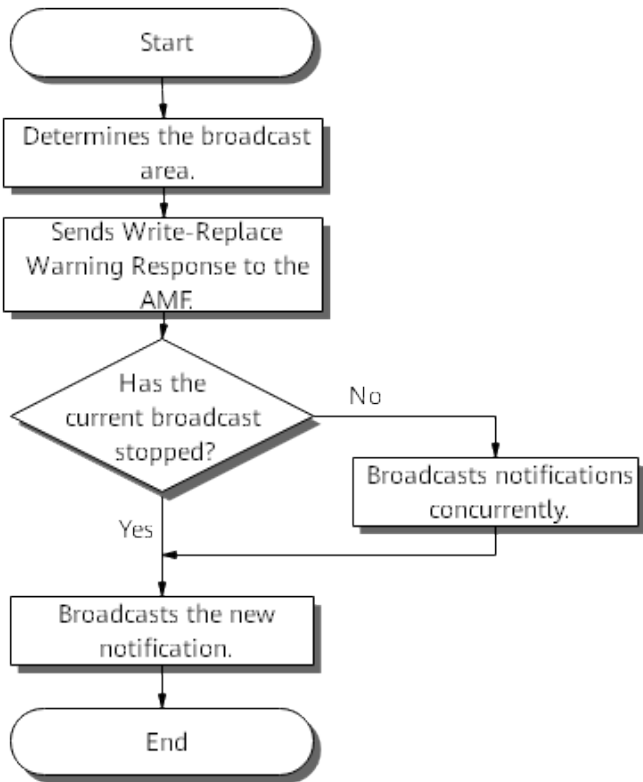
7. The gNodeB sends paging indication to the UE located in the broadcast area and delivers SIB messages. The gNodeB delivers SIB8 during paging. To improve system reliability, the gNodeB supports supplementary SIB8 transmission. This function is controlled by the **SIB8_SUPP_TRANS_SW** option of the **NRDUCellAlgoSwitch.AlgoCompatibilitySwitchExt** parameter. If this option is selected, the gNodeB sends supplementary SIB8 messages to enhance reliability after SIB8 transmission is complete. In addition, the time for broadcasting SIB8 over the air interface is prolonged. If this option is deselected, supplementary transmission is not performed after SIB8 transmission is complete.
8. The UE receives the CMAS broadcast message and notifies the user of the message.

4.1.3.2 gNodeB Handling Procedure

Overview

The gNodeB handling procedure corresponds to step 6 in the overall procedure. [Figure 4-3](#) shows the gNodeB handling procedure after the gNodeB receives a Write-Replace Warning Request message from the AMF in step 5.

Figure 4-3 gNodeB handling procedure



NOTE

If the broadcast of a warning notification in a cell is not complete and the cell is restarted due to certain reasons, the gNodeB clears this notification and does not deliver it again after the cell restart.

Determining a Broadcast Area

Table 4-1 Determining a broadcast area

If...	Then...
The Write-Replace Warning Request message does not contain the <i>Warning Area List</i> IE, and the CMAS_BROADCAST_OP_FILTER_SW option of the gNodeBAlgo.RanSharingAlgoSw parameter is deselected	All available cells served by the gNodeB are included in the broadcast area.
The Write-Replace Warning Request message does not contain the <i>Warning Area List</i> IE, and the CMAS_BROADCAST_OP_FILTER_SW option of the gNodeBAlgo.RanSharingAlgoSw parameter is selected	The gNodeB treats all available cells of the operators corresponding to the core network that delivers CMAS warning notifications as the broadcast area.

If...	Then...
The Write-Replace Warning Request message contains the <i>Warning Area List</i> IE and this IE indicates that the warning notification is broadcast based on the NR CGI list	The gNodeB maps the NR CGIs to the corresponding cells if available and only mapped cells are included in the broadcast area.
The Write-Replace Warning Request message contains the <i>Warning Area List</i> IE and this IE indicates that the warning notification is broadcast based on the TAI list	The gNodeB maps the TAIs to the corresponding cells if available and only mapped cells are included in the broadcast area.
The Write-Replace Warning Request message contains the <i>Warning Area List</i> IE, but the <i>Warning Area List</i> IE is inconsistent with that on the gNodeB side	No broadcast area exists and the gNodeB does not broadcast warning notifications. In this case, the gNodeB responds to the AMF with a Write-Replace Warning Response message not containing the <i>Broadcast Completed Area List</i> IE. Upon receiving this message, the AMF considers that the broadcast failed. The AMF then notifies the CBE of the failure through the CBCF/PWS-IWF. The CBE determines the subsequent processing policy as planned.

Broadcasting Warning Notifications

Table 4-2 describes the warning notification broadcast policies.

Table 4-2 Warning notification broadcast policies

Repetition Period (s)	Number of Broadcasts Requested	Broadcast Policy of a Warning Notification
= 0	= 1	The gNodeB broadcasts the warning notification only once.
	≠ 1	The gNodeB does not broadcast the warning notification.
> 0	> 0	The gNodeB broadcasts the warning notification at an interval specified by <i>Repetition Period</i> . The number of broadcasts is specified by <i>Number of Broadcasts Requested</i> .

Repetition Period (s)	Number of Broadcasts Requested	Broadcast Policy of a Warning Notification
	= 0	The gNodeB broadcasts the warning notification based on <i>Repetition Period</i> until it receives the stop request.

 NOTE

Both *Repetition Period* and *Number of Broadcasts Requested* are contained in the Write-Replace Warning Request message. The *Repetition Period* IE specifies the interval at which a warning notification is broadcast, in the unit of seconds.

Concurrent Delivery of Warning Notifications

Table 4-3 Concurrent delivery of warning notifications

If...	Then...
The new Write-Replace Warning Request message contains a different <i>Message Identifier</i> or <i>Serial Number</i> from the warning notification being broadcast	The gNodeB regards the newly received Write-Replace Warning Request message as a different warning request, regardless of whether the broadcast areas are the same. The gNodeB broadcasts the warning notifications within the broadcast area specified by the Write-Replace Warning Request message without interrupting the ongoing warning notification broadcast. In this situation, concurrent delivery of warning notifications is required.
The new Write-Replace Warning Request message contains the same <i>Message Identifier</i> and <i>Serial Number</i> as the warning notification being broadcast, and their broadcast areas are also the same	The gNodeB discards the newly received warning notification without interrupting the ongoing warning notification broadcast.
The new Write-Replace Warning Request message contains the same <i>Message Identifier</i> and <i>Serial Number</i> as the warning notification being broadcast, and their broadcast areas are different	The gNodeB broadcasts the new warning notification in the new broadcast areas without interrupting the ongoing warning notification broadcast. In this situation, concurrent delivery of warning notifications is required.

The following warning notifications are to be delivered concurrently:

- Warning notifications that are newly received by the gNodeB and have not been broadcast yet
- Warning notifications to be retransmitted, namely, warning notifications that have been broadcast by the gNodeB at least once, but the number of broadcasts has not reached the value specified by *Number of Broadcasts Requested*.

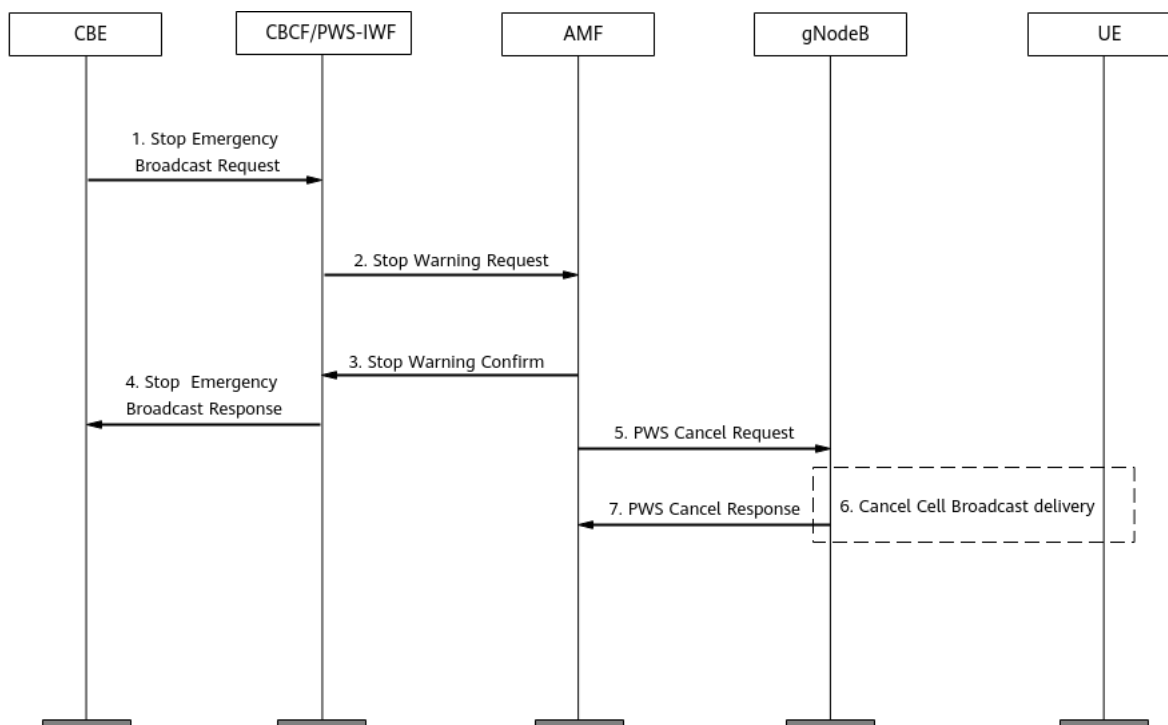
The gNodeB can concurrently deliver a maximum of 64 warning notifications. The warning notifications are broadcast according to the following priorities:

- Newly received warning notifications that have not been broadcast have a higher priority than warning notifications to be retransmitted.
- If there are multiple newly received warning notifications that have not been broadcast, the earliest received warning notification is preferentially delivered.
- If there are multiple to-be-retransmitted warning notifications, the warning notification whose waiting time is the longest since its last transmission is preferentially delivered.

4.1.4 Broadcast Stop Procedure

4.1.4.1 Overall Procedure

Figure 4-4 Warning notification broadcast stop procedure



1. The CBE sends a Stop Emergency Broadcast Request message to the CBCF/PWS-IWF. The request contains the warning notification and broadcast areas where the warning notification broadcast is to be stopped.

2. The CBCF/PWS-IWF converts the received message into a standardized signaling message Stop Warning Request defined by 3GPP TS 23.041 and then sends the message to the AMF.
3. The AMF responds to the CBCF/PWS-IWF with a Stop Warning Confirm message, declaring that the message for stopping broadcasting has been forwarded to the gNodeB.
4. The CBCF/PWS-IWF sends a Stop Emergency Broadcast Response message to the CBE, declaring that the warning notification stop procedure has been started.
5. The AMF sends a PWS Cancel Request message to the gNodeB located in the broadcast areas indicated in the Stop Warning Request message.
6. The gNodeB stops broadcasting warning notifications based on the PWS Cancel Request message.
7. The gNodeB sends a PWS Cancel Response message to the AMF, notifying the AMF of the areas where broadcasting of the warning notification has been stopped.

4.1.4.2 gNodeB Handling Procedure

If...	Then...
The served cells of the gNodeB belong to the areas where the warning notification broadcast is to be stopped, and the <i>Message Identifier</i> and <i>Serial Number</i> IEs contained in the PWS Cancel Request message are the same as those in the warning notification being broadcast	The gNodeB stops broadcasting the warning notification, stops sending paging messages for it, and cancels it in broadcast areas indicated by the <i>Warning Area List</i> IE.
The served cells of the gNodeB belong to the areas where the warning notification broadcast is to be stopped, and the <i>Message Identifier</i> and <i>Serial Number</i> IEs contained in the PWS Cancel Request message are different from those in the warning notification being broadcast	The gNodeB does not perform any operation.
The served cells of the gNodeB do not belong to the areas where the warning notification broadcast is to be stopped	The gNodeB does not perform any operation in response to the PWS Cancel Request message.

If...	Then...
The PWS Cancel Request message does not contain the <i>Warning Area List</i> IE	The gNodeB stops broadcasting the warning notification with the same <i>Message Identifier</i> and <i>Serial Number</i> IEs in the PWS Cancel Request message, stops sending paging messages for the warning notification, and cancels the warning notification in all cells served by the gNodeB, regardless of whether the CMAS_BROADCAST_OP_FILTER_SW option of the gNodeBAlgo.RanSharingAlgoSw parameter is selected or deselected.
The <i>Warning Area List</i> IE contained in the PWS Cancel Request message indicates that the cancellation is performed based on the NR CGI list	The gNodeB stops broadcasting the warning notification with the same <i>Message Identifier</i> and <i>Serial Number</i> IEs in the PWS Cancel Request message, stops sending paging messages for the warning notification, and cancels the warning notification in the cells indicated in the NR CGI list.
The <i>Warning Area List</i> IE contained in the PWS Cancel Request message indicates that the cancellation is performed based on the TAI list	The gNodeB stops broadcasting the warning notification with the same <i>Message Identifier</i> and <i>Serial Number</i> IEs in the PWS Cancel Request message, stops sending paging messages for the warning notification, and cancels the warning notification in the cells indicated in the TAI list.
The PWS Cancel Request message contains the <i>Cancel-all Warning Message Indicator</i> and <i>Warning Area List</i> IEs	The gNodeB stops broadcasting in broadcast areas indicated by the <i>Warning Area List</i> IE and cancels all warning notifications. In this case, the <i>Message Identifier</i> and <i>Serial Number</i> IEs do not need to be judged.
The PWS Cancel Request message contains the <i>Cancel-all Warning Message Indicator</i> IE but does not contain the <i>Warning Area List</i> IE	Regardless of whether the CMAS_BROADCAST_OP_FILTER_SW option of the gNodeBAlgo.RanSharingAlgoSw parameter is selected or deselected, the gNodeB stops broadcasting warning notifications to all cells served by the gNodeB and cancels all warning notifications.

4.1.5 PWS Restart Indication Procedure

The public warning system (PWS) restart indication procedure is controlled by the **PWS_RESTART_IND_SW** option of the **gNodeBParam.PwsIndicationSwitch** parameter. If this option is selected, the gNodeB sends a PWS Restart Indication message to the AMF after a served cell is restarted (that is, the cell status changes from unavailable to available).

- If the gNodeB is connected to only one AMF, the gNodeB directly sends the PWS Restart Indication message to the AMF.
- If the gNodeB is connected to multiple AMFs, the gNodeB sends the PWS Restart Indication message to all available AMFs to ensure that the message can be sent to the CBE even if an AMF is faulty.

4.1.6 PWS Failure Indication Procedure

The PWS failure indication procedure is controlled by the **PWS_FAILURE_IND_SW** option of the **gNodeBParam.PwsIndicationSwitch** parameter. If this option is selected, the gNodeB sends a PWS Failure Indication message to the AMF when the status of a cell served by the gNodeB changes from available to unavailable and a PWS message is being sent in the cell.

- If the gNodeB is connected to only one AMF, the gNodeB directly sends the PWS Failure Indication message to the AMF.
- If the gNodeB is connected to multiple AMFs, the gNodeB sends the PWS Failure Indication message to all available AMFs to ensure that the message can be sent to the CBE even if an AMF is faulty.

4.2 Network Analysis

4.2.1 Benefits

- CMAS is recommended when broadcasting warning notifications about emergencies to the public through PLMNs is required.
- With CMAS, users can be notified of imminent dangers or threats promptly, helping prevent possible damages.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

- When CMAS is enabled, the number of configured system information messages needs to be changed from 4 to 5, and the periods for SIB2 to SIB5 need to be changed from 320 ms to 640 ms. As a result, the average number of PDCCH CCEs used for common DCI (**N.CCE.CommDCI.Used.Avg**) decreases, the average number of available downlink PRBs excluding those for common signaling (**N.PR.B.DL.RmvCommSig.Avail.Avg**) increases, and the contention-based access success rate (**N.RA.Contention.Resolution.Succ/N.RA.Contention.Att**) and non-contention-based access success rate (**N.RA.Dedicated.Msg3/N.RA.Dedicated.Att**) slightly fluctuate.

- Broadcasting CMAS warning notifications over the air interface occupies air interface resources, affecting the downlink cell throughput. The extent of the impact varies with the number of CMAS warning notification fragments, the number of times CMAS warning notifications are broadcast, and the broadcast repetition period.
- Enabling supplementary SIB8 transmission occupies more air interface resources, which in turn affects the downlink cell throughput. The extent of the impact varies with the number of CMAS warning notification fragments, the number of times CMAS warning notifications are broadcast, and the broadcast repetition period.
- The number of cells in which CMAS warning notifications are broadcast decreases, UEs in cells without CMAS warning notification broadcast do not receive CMAS warning notifications, and the downlink PRB usage decreases if the following conditions apply: (1) the **CMAS_BROADCAST_OP_FILTER_SW** option of the **gNodeBAlgo.RanSharingAlgoSw** parameter is selected; (2) the base station works in RAN sharing with dedicated carrier or RAN sharing with common carrier mode; (3) the CMAS warning notifications delivered by the core network do not contain the *Warning Area List* IE.

4.3 Requirements

4.3.1 Licenses

There are no license requirements.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
System information broadcast	None	<i>5G Networking and Signaling</i>	None

Function Name	Function Switch	Reference	Description
CMAS	CMAS_SWITCH option of the NRDUCellFeatureSw. PwsSwitch parameter	<i>CMAS</i>	When a cell is activated, if the gNBSibConfig.SibType parameter corresponding to the SibConfig MO referenced by the NRDUCell MO is set to SIB_TYPE_8 and the gNBSibConfig.SibTransPolicy parameter is set to BROADCAST , the CMAS_SWITCH option of the NRDUCellFeatureSw. PwsSwitch parameter must be selected.

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	LTE and NR intelligent carrier shutdown	LNR_SMART_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When LTE and NR intelligent carrier shutdown takes effect, CMAS warning notifications cannot be sent in NR cells since those cells are blocked. However, UEs can still receive CMAS warning notifications from other carriers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARRER_SD_SW or INTER_GNB_MULTI_CARRER_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown takes effect, CMAS warning notifications cannot be sent in NR cells since those cells are blocked. However, UEs can still receive CMAS warning notifications from other carriers.
Low-frequency TDD	Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When power saving scheduling takes effect, the interval at which system information is sent changes. The base station updates the paging occasion after two paging periods elapse. However, a UE updates the paging occasion to receive paging messages only after receiving SIB1. Therefore, there is a low probability that the UE fails to receive CMAS warning notifications in this scenario. CMAS warning notifications are usually sent multiple times by the CBCF/PWS-IWF. This ensures that UEs can receive CMAS warning notifications in power saving scheduling scenarios.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When timing carrier shutdown takes effect, CMAS warning notifications cannot be sent in NR cells since those cells are blocked. However, UEs can still receive CMAS warning notifications from other carriers.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW (LTE eNodeB, 5G gNodeB) set to ON or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	When RF module deep dormancy takes effect, CMAS warning notifications cannot be sent in NR cells since those cells are blocked. However, UEs can still receive CMAS warning notifications from other cells.
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (5G gNodeB, LTE eNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When RF module fast deep dormancy takes effect, CMAS warning notifications cannot be sent in NR cells since those cells are blocked. However, UEs can still receive CMAS warning notifications from other carriers.
Low-frequency TDD	RF module super dormancy	RRU.DORMANCYSW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	When RF module super dormancy takes effect, CMAS warning notifications cannot be sent in NR cells since those cells are blocked. However, UEs can still receive CMAS warning notifications from other cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- Capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter (intra-base-station multi-carrier scenarios) or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter (inter-base-station multi-carrier scenarios) FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter - Basic-layer cells:FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When fast carrier shutdown takes effect, CMAS warning notifications cannot be sent in NR cells since those cells are blocked. However, UEs can still receive CMAS warning notifications from other carriers.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

In low-frequency scenarios, base station models must meet the following requirements:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

N/A

4.3.4 Cells

CMAS is not supported in SDL cells because the cells do not send OSI. **NRDUCell.DuplexMode** cannot be set to **CELL_SDL**.

When the **NRDUCell.SsbPeriod** parameter is set to **MS5** or **MS10** for a cell, the time-domain positions of SSB and OSI may conflict with each other. As a result, CMAS warning notifications fail to be sent. It is recommended that the **NRDUCell.SsbPeriod** parameter not be set to **MS5** or **MS10** for a cell.

4.3.5 Others

Deploying this feature requires the following:

- The AMF, CBCF/PWS-IWF, and UE support the CMAS feature.
- The same broadcast areas are defined in CBEs and the PLMN.
- The AMF complies with 3GPP Release 16 or later and the AMF protocol version of the NG interface on the gNodeB side is consistent with that on the core network.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Parameter Name	Parameter ID	Option	Setting Notes
SIB Config ID	gNBSibConfig.SibConfigId	None	Set this parameter based on site requirements.

Parameter Name	Parameter ID	Option	Setting Notes
SIB Config ID	NRDUCell.SibConfigId	None	Set this parameter based on site requirements.
SIB Type	gNBSibConfig.SibType	None	Set this parameter based on site requirements.
SIB Transmission Policy	gNBSibConfig.SibTransPolicy	None	Set this parameter to BROADCAST .
SIB Transmission Period	gNBSibConfig.SibTransPeriod	None	Set this parameter based on site requirements.
PWS Indication Switch	gNodeBParam.PwsIndicationSwitch	PWS_RESTART_IND_SW	To turn on the PWS restart indication switch, select this option.
PWS Indication Switch	gNodeBParam.PwsIndicationSwitch	PWS_FAILURE_IND_SW	To turn on the PWS failure indication switch, select this option.
CMAS Broadcast Switch	gNBOperator.CmasBroadcastSwitch	None	This parameter is set to ON by default. You are advised to set this parameter to ON in scenarios where the multi-operator core network (MOCN) architecture is used, the CBCF/PWS-IWF are independently deployed and do not work in a coordinating way, and the base station works in RAN sharing with common carrier or hybrid RAN sharing modes. Set the gNBOperator.CmasBroadcastSwitch parameter to OFF for a PLMN from which CMAS warning notifications do not need to be broadcast.

Parameter Name	Parameter ID	Option	Setting Notes
Public Warning System Switch	NRDUCellFeatureSw.PwsSwitch	CMAS_SWITCH	Select this option if CMAS is required.
Algorithm Compatibility Switch Extension	NRDUCellAlgoSwitch.AlgoCompatibilitySwitchExt	SIB8_SUPP_TRANS_SW	It is recommended that this option be selected when system reliability needs to be improved.
RAN Sharing Algorithm Switch	gNodeBAlgo.RANSharingAlgoSw	CMAS_BROADCAST_OP_FILTER_SW	Select this option when broadcast range control of CMAS warning notifications based on operators is required.

 NOTE

For a system information configuration ID, if the minimum system information transmission period is 320 ms, a maximum of four system information types can be configured. If the minimum system information transmission period is 640 ms, a maximum of eight system information types can be configured.

4.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact and dependency relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

- If a cell uses the default system information configuration (that is, a set of system information configuration whose ID is 0), a new set of system information configuration needs to be created and bound to the cell when the default system information configuration cannot meet the service requirements. This happens because the default system information configuration cannot be modified.
- If a cell uses a non-default system information configuration (for example, a set of system information configuration whose ID is 1), the system information type being set to SIB8 must be added to activate CMAS. In this case, the system information transmission period is prolonged and it takes a longer time for UEs to obtain such system information than the default system information.

```
//Adding gNodeB system information configuration
ADD GNBSIBCONFIG: SibConfigId=1, SibType=SIB_TYPE_8, SibTransPeriod=640MS,
SibTransPolicy=BROADCAST;
ADD GNBSIBCONFIG: SibConfigId=1, SibType=SIB_TYPE_2, SibTransPeriod=640MS,
SibTransPolicy=BROADCAST;
ADD GNBSIBCONFIG: SibConfigId=1, SibType=SIB_TYPE_3, SibTransPeriod=640MS,
SibTransPolicy=BROADCAST;
ADD GNBSIBCONFIG: SibConfigId=1, SibType=SIB_TYPE_4, SibTransPeriod=640MS,
SibTransPolicy=BROADCAST;
ADD GNBSIBCONFIG: SibConfigId=1, SibType=SIB_TYPE_5, SibTransPeriod=640MS,
SibTransPolicy=BROADCAST;
//Turning on the CMAS switch for a cell
MOD NRDUCELLFEATURESW: NrDuCellId=0, PwsSwitch=CMAS_SWITCH-1;
//Modifying the system information configuration of a cell
MOD NRDUCELL: NrDuCellId=0, SibConfigId=1;
//(Optional) Turning on the PWS restart indication switch
MOD GNODEBPARAM: PwsIndicationSwitch=PWS_RESTART_IND_SW-1;
//(Optional) Turning on the PWS failure indication switch
MOD GNODEBPARAM: PwsIndicationSwitch=PWS_FAILURE_IND_SW-1;
//(Optional) Turning off the CMAS broadcast switch for an operator in the MOCN scenario
MOD GNOPERATOR: OperatorId=0, CmasBroadcastSwitch=OFF;
//(Optional) Turning on the supplementary SIB8 transmission switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AlgoCompatibilitySwitchExt=SIB8_SUPP_TRANS_SW-1;
//(Optional) Turning on the switch controlling operator filtering for CMAS broadcast
MOD GNODEBALGO: RanSharingAlgoSw=CMAS_BROADCAST_OP_FILTER_SW-1;
```

Deactivation Command Examples

```
//Modifying the gNodeB system information configuration to deactivate the CMAS function:
MOD GNBSIBCONFIG: SibConfigId=1, SibType=SIB_TYPE_8, SibTransPeriod=640MS, SibTransPolicy=OFF;
//Turning off the CMAS switch for a cell
MOD NRDUCELLFEATURESW: NrDuCellId=0, PwsSwitch=CMAS_SWITCH-0;
//(Optional) Turning off the PWS restart indication switch
MOD GNODEBPARAM: PwsIndicationSwitch=PWS_RESTART_IND_SW-0;
//(Optional) Turning off the PWS failure indication switch
MOD GNODEBPARAM: PwsIndicationSwitch=PWS_FAILURE_IND_SW-0&PWS_FAILURE_IND_ENH_SW-0;
//(Optional) Turning off the supplementary SIB8 transmission switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AlgoCompatibilitySwitchExt=SIB8_SUPP_TRANS_SW-0;
//(Optional) Turning off the switch controlling operator filtering for CMAS broadcast
MOD GNODEBALGO: RanSharingAlgoSw=CMAS_BROADCAST_OP_FILTER_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

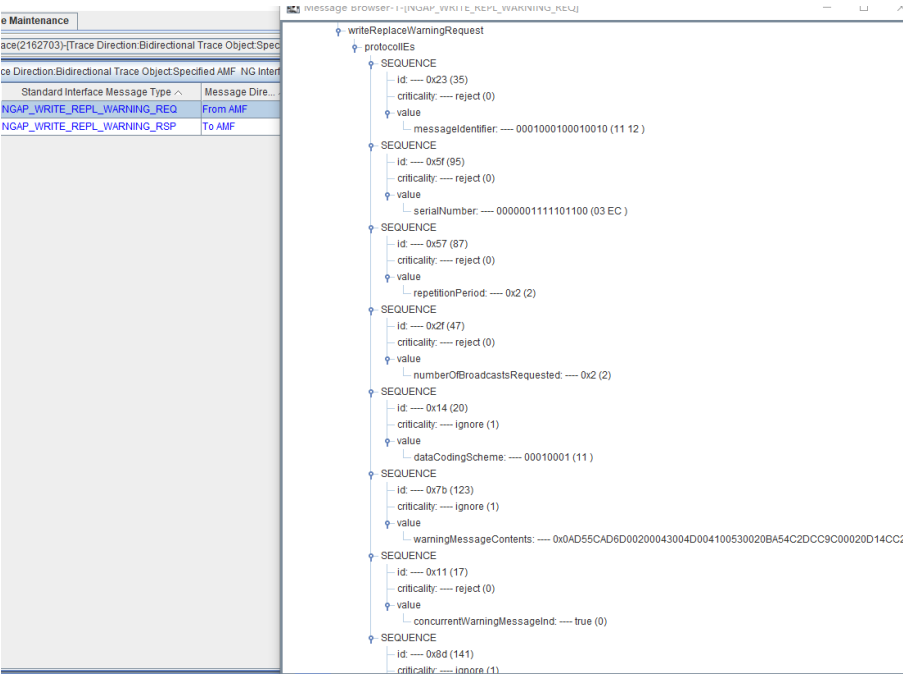
If CMAS-related messages are sent normally over the NG and Uu interfaces, CMAS has been activated. To trace messages over the NG and Uu interfaces, perform the following steps on the MAE-Access:

- Step 1** Enable the core network to send a Write-Replace Warning Request message to gNodeBs.
- Step 2** Trace the NG messages sent to and from all gNodeBs located in the broadcast areas, which are indicated in the Write-Replace Warning Request message.

If the Write-Replace Warning Request from the core network and the Write-Replace Warning Response message from the gNodeB can be traced, the message

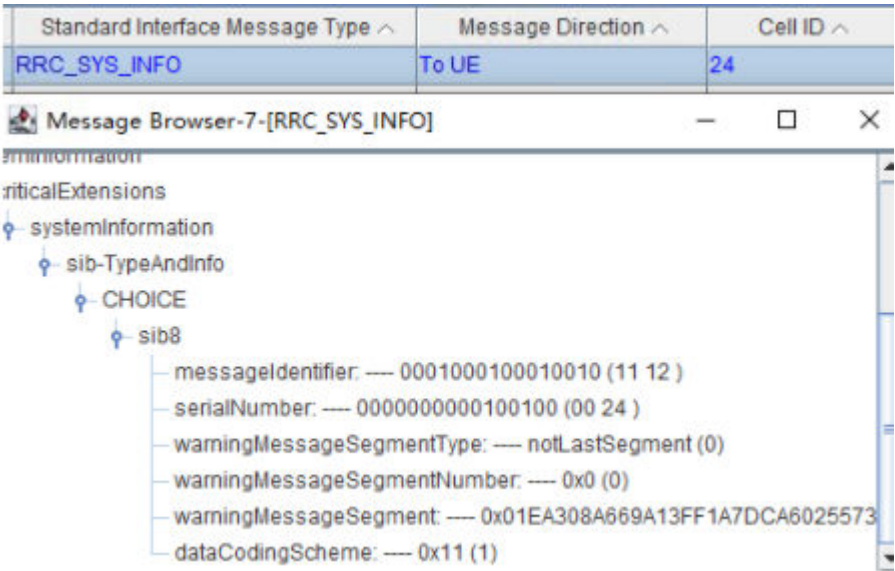
exchange over the NG interfaces is normal. **Figure 4-5** shows the messages over the NG interface.

Figure 4-5 NG interface tracing result



Step 3 Trace and observe messages over the Uu interface to check the system information delivered by the gNodeB. Double-click **RRC_SYS_INFO**. If SIB8 contains a CMAS warning notification, the message exchange over the Uu interface is normal. **Figure 4-6** shows Uu messages.

Figure 4-6 Uu interface tracing result



----End

4.4.3 Network Monitoring

The following counters are used to observe the number of times WRITE-REPLACE WARNING REQUEST messages are received by the gNodeB, number of times PWS CANCEL REQUEST messages are received by the gNodeB, number of successful CMAS broadcast times, number of times PWS RESTART INDICATION messages are sent to the core network, and number of times PWS FAILURE INDICATION messages are sent to the core network.

Counter ID	Counter Name
1911827200	N.Signal.WRWR.Num
1911827201	N.Signal.PWS.CancelReq.Num
1911829346	N.Signal.CMAS.Broadcast.Succ
1911829347	N.Signal.PWS.RestartIndication.Num
1911829348	N.Signal.PWS.FailureIndication.Num

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR;Base Station (BS) radio transmission and reception"
- 3GPP TS 38.413
- 3GPP TS 23.041
- *Multi-RAT Coordinated Energy Saving*
- *Energy Conservation and Emission Reduction*
- *5G Networking and Signaling*